

Discrete Time Signal Processing

Applications of Digital Signal Processing

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Application of DSP for ECG signals analysis



Electrocardiogram (ECG)

- What is ECG?

Diagnostic tool that graphically measures and records electrical activity of heart in detail.

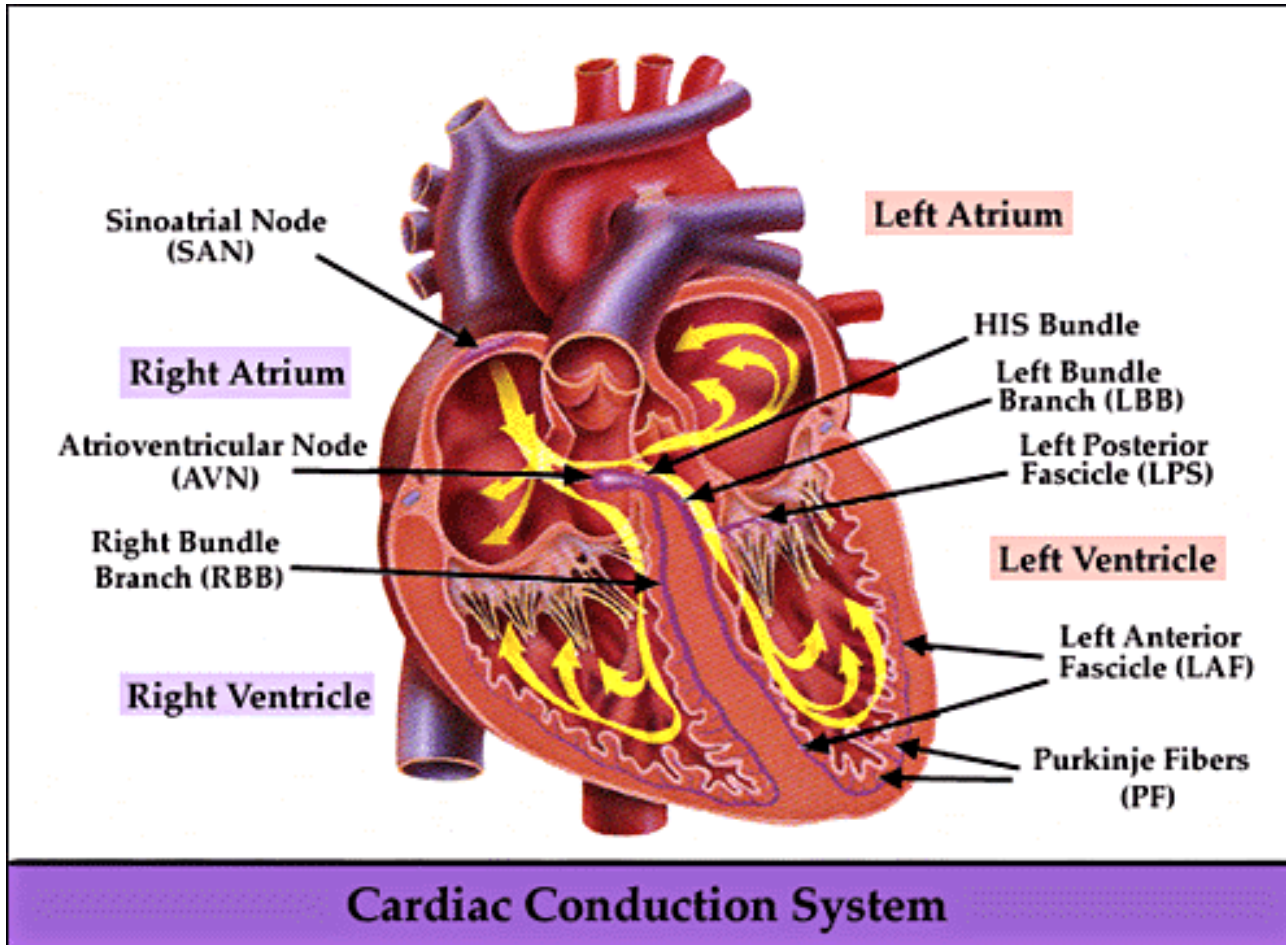
- One of the main methods for assessing heart functions.

- Many cardiac parameters, such as heart rate, myocardial infarction, and enlargement can be determined.

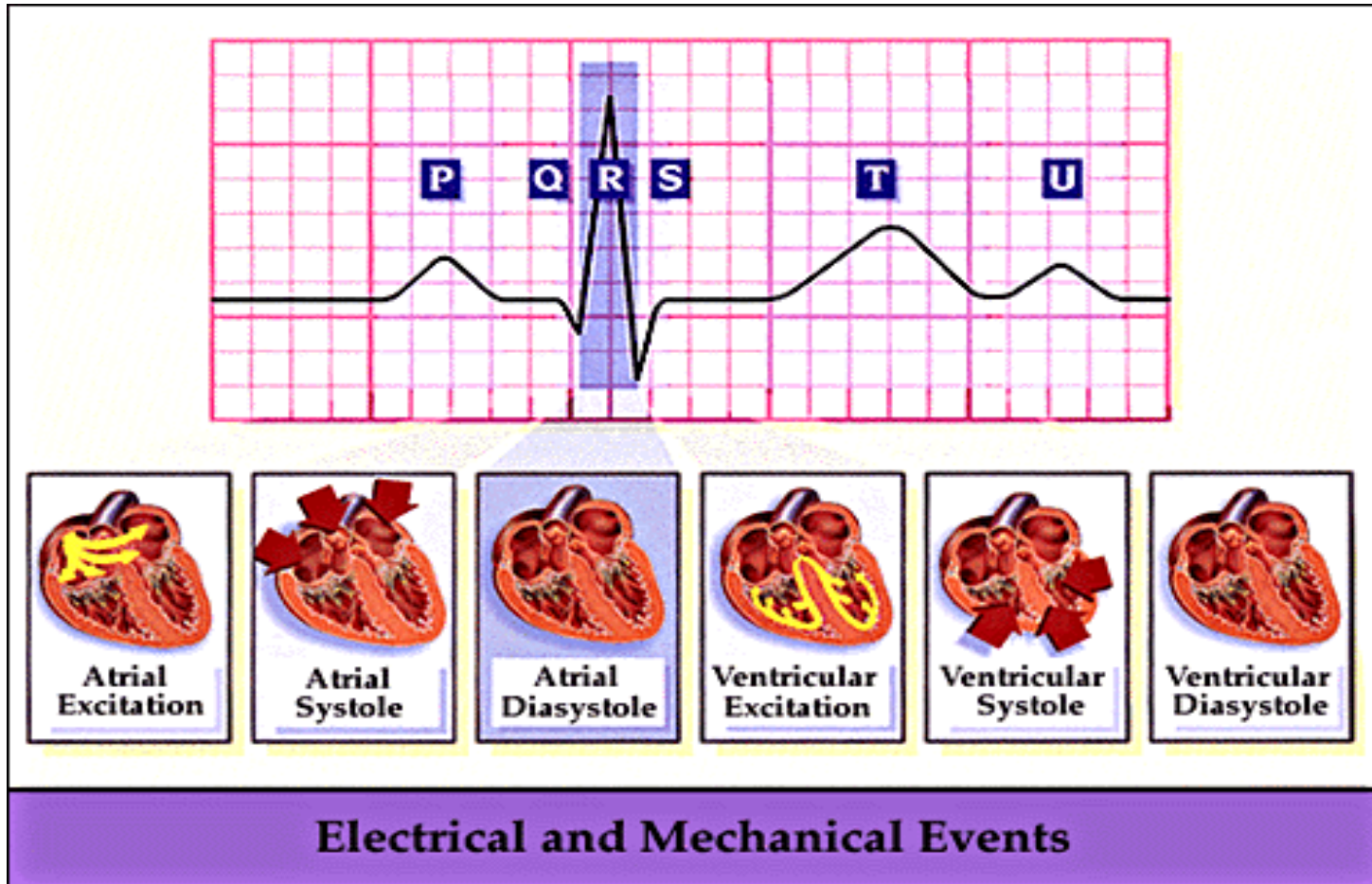
- The potential difference between two electrodes placed on skin surface is considered as an input to ECG plotter.

- A single normal cycle of ECG represents successive atrial depolarization/ repolarization and ventricular depolarization/ repolarization which occurs with every heartbeat.

Application of DSP for ECG signals analysis



Application of DSP for ECG signals analysis



ECG Diagnosis

- Following pulses that completes ECG wave:
 - 1.P-wave,
 - 2.QRS complex,
 - 3.T-wave.
 - 4.U- wave
- A single normal cycle of ECG represents successive artial depolarization/ repolarization and ventricular depolarization/ repolarization which occurs with every heartbeat.
- P-wave- sequential activation(depolarization) of right and left atria.
- QRS complex- right and left ventricular depolarization
- T-wave- ventricular repolarization
- U-wave- Origin not clear, probably after depolarization in the ventricles.

Elements of ECG

P- wave

- depolarization of both atria
- relationship between P & QRS helps distinguish various cardiac arrhythmias.
- shape and duration of P may indicate atrial enlargement.

QRS complex

- represents ventricular depolarization
- larger than P-wave because of greater muscle mass of ventricles
- Normal duration 0.08-0.12second
- Its duration, amplitude & morphology are useful in diagnosing cardiac arrhythmias, ventricular hypertrophy, MI, electrolyte derangement etc.
- Q wave greater than 0.04second are abnormal and represent MI.

Elements of ECG

PR interval :

- from onset of P-wave to onset of QRS complex
- normal duration- 0.12 to 0.2sec
- represents atria to ventricular conduction time (through His bundle)
- prolonged PR interval may indicate a 1st degree block

ST segment

- connects QRS complex & T wave
- duration of 0.08 to 0.12sec

Elements of ECG

T-wave

- Represents repolarization or recovery of ventricles
- Interval from beginning of QRS to apex of T is referred as absolute refractory period

QT interval

- Measured from beginning of QRS to end of T
- normal QT is usually 0.40sec
- QT interval varies based on beat rate.

Application of DSP for ECG signals analysis

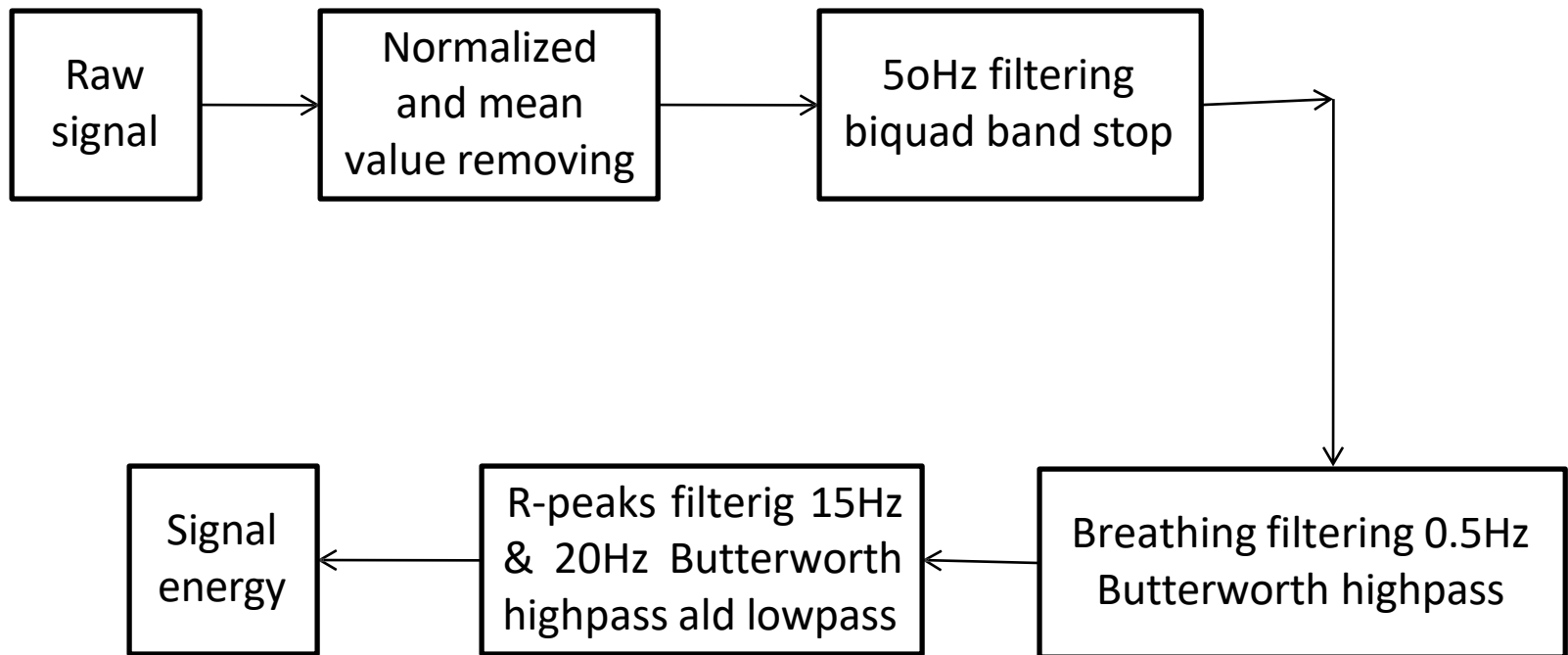
Need of using DSP

- processing performance at very low power
- main problem of digital signal is interference with other noisy signals like power supply network 50Hz frequency breathing muscle artifats.
- these noisy elements have to be removed before signal is used for next data processing like heart rate frequency detection.

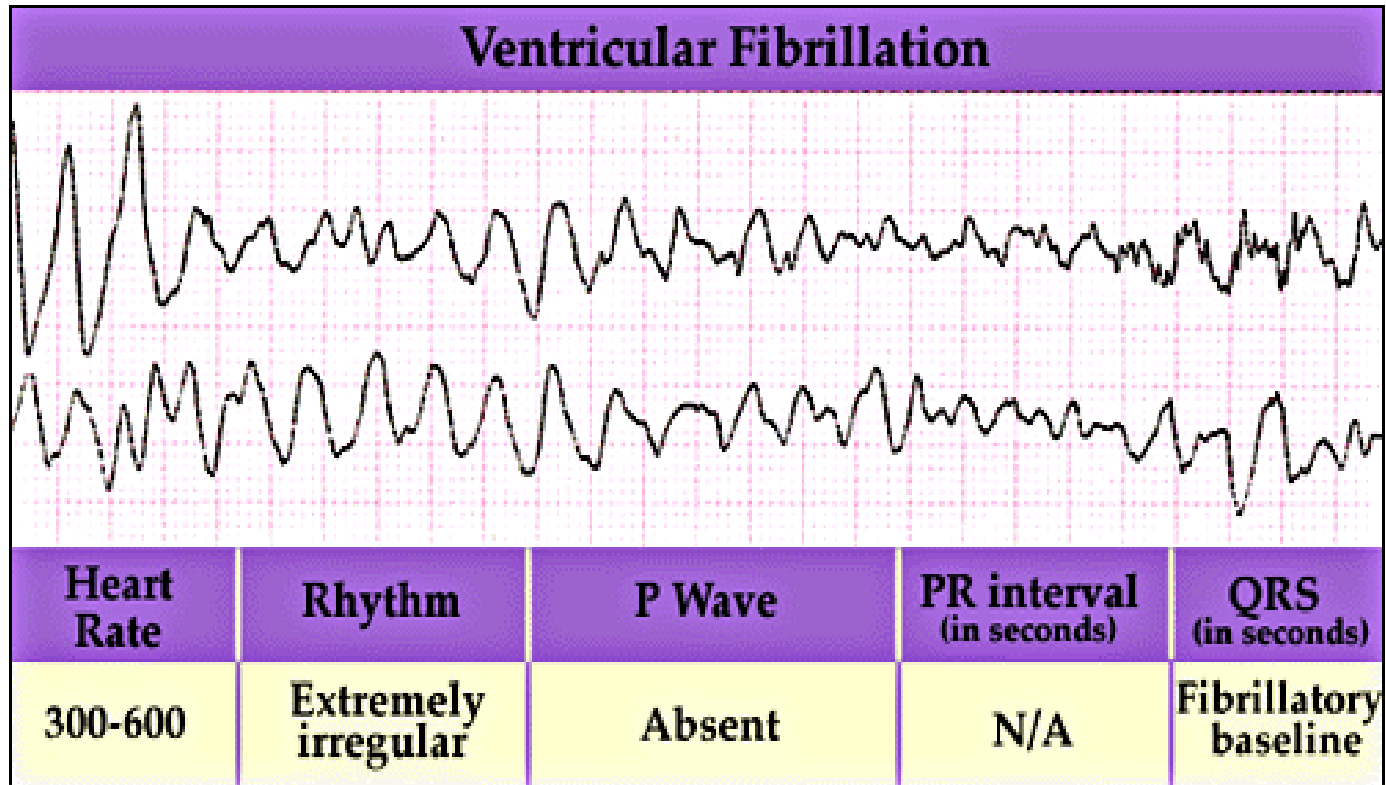
Steps involved :

- signal acquisition
- signal processing with digital filter
- heart rate decision algorithms of two types
 1. statistical computing
 2. differential computing

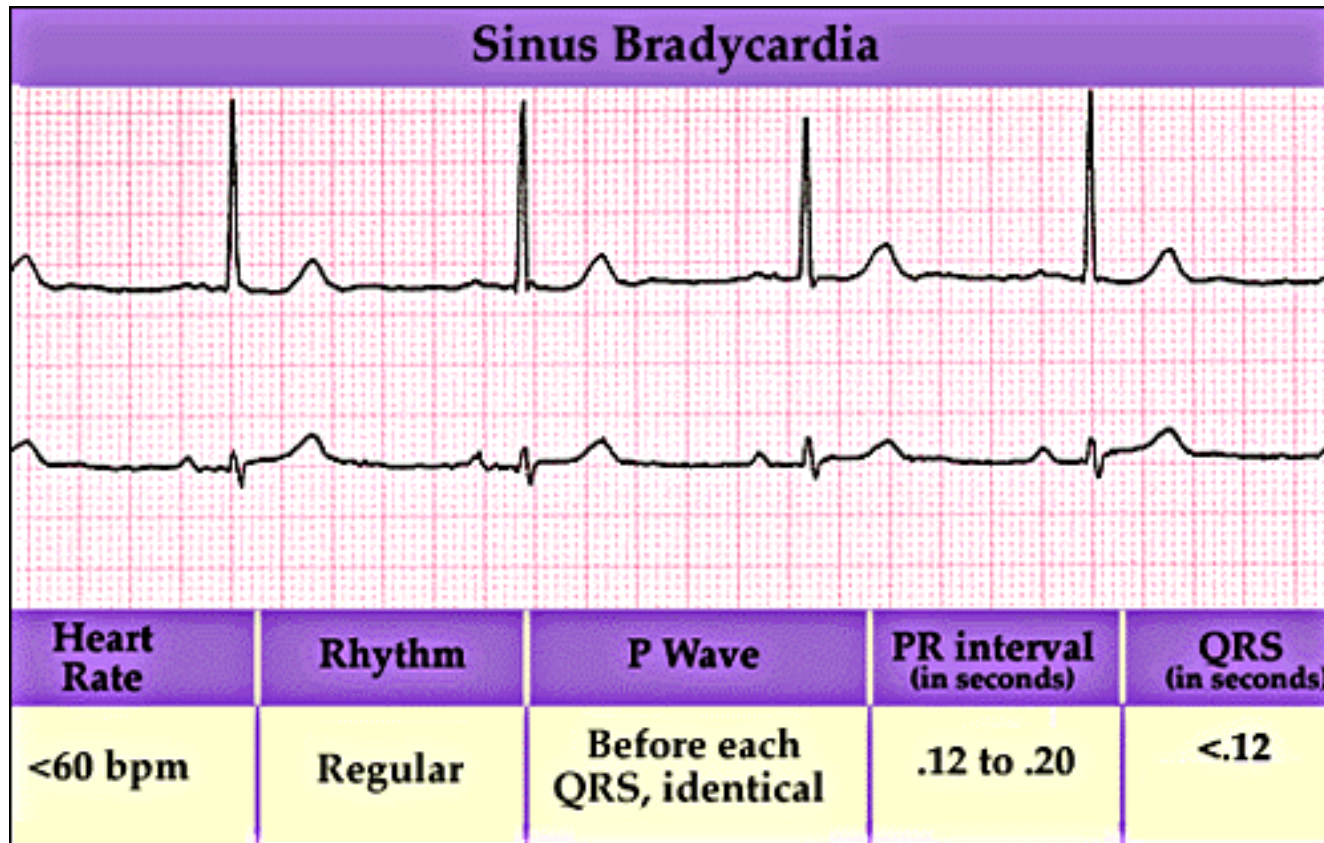
Application of DSP for ECG signals analysis



ECG Diagnosis



ECG Diagnosis



Application of DSP for Dual Tone Multi Frequency signal detection



Application of DSP for Dual Tone Multi Frequency signal detection

- Dual Tone Multi-Frequency, or DTMF is a method for instructing a telephone switching system of the telephone number to be dialed, or to issue commands to switching systems or related telephony equipment.
- The DTMF dialing system traces its roots to a technique AT&T developed in the 1950s called MF (Multi-Frequency) which was deployed within the AT&T telephone network to direct calls between switching facilities using in-band signaling.
- The frequencies used were chosen to prevent any harmonics from being incorrectly detected by the receiver as some other DTMF frequency.
- The transmitter of a DTMF signal simultaneously sends one frequency from the high-group and one frequency from the low-group. This pair of signals represents the digit or symbol shown at the intersection of row and column in the table. For example, sending 1209Hz and 770Hz indicates that the "4" digit is being sent.

Application of DSP for Dual Tone Multi Frequency signal detection

- At the transmitter, the maximum signal strength of a pair of tones must not exceed +1 dBm, and the minimum strength is -10.5 dBm for the low-group frequencies and -8.5 dBm for the high-group frequencies.
- The DTMF system uses eight different frequency signals transmitted in pairs to represent sixteen different numbers, symbols and letters. This table shows how the frequencies are organized:

		High-Group Frequencies			
		1209Hz	1336Hz	1477Hz	1633Hz
Low-Group Frequencies	697Hz	1	ABC 2	DEF 3	A
	770Hz	GHI 4	JKL 5	MNO 6	B
	852Hz	PRS 7	TUV 8	WXY 9	C
	941Hz	*	OPER 0	#	D

Application of DSP for Dual Tone Multi Frequency signal detection

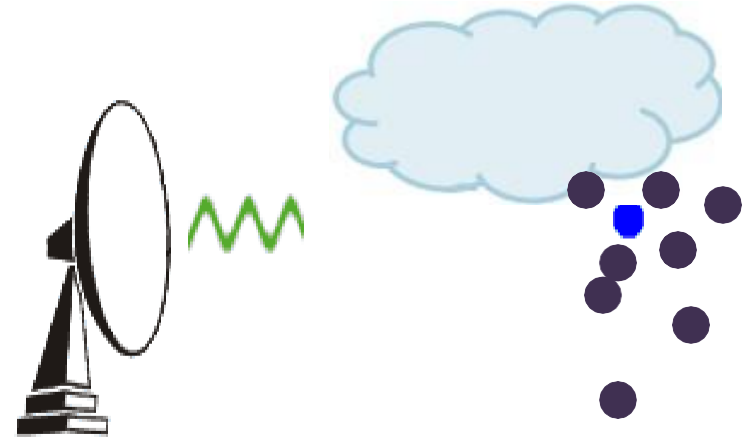
- Labeling of DTMF numeric digits :
- The DTMF telephone keypad is laid out in a 4×4 matrix of push buttons in which each row represents the low frequency component and each column represents the high frequency component of the DTMF signal. Pressing a key sends a combination of the row and column frequencies.
- For example, the key 1 produces a superimposition of tones of 697 and 1209 hertz (Hz). Initial pushbutton designs employed levers, so that each button activated two contacts. The tones are decoded by the switching center to determine the keys pressed by the user..
- DTMF was originally decoded by tuned filter banks. By the end of the 20th century, digital signal processing became the predominant technology for decoding. DTMF decoding algorithms often use the Goertzel algorithm to detect tones.

Application of DSP for Radar Signal Processing

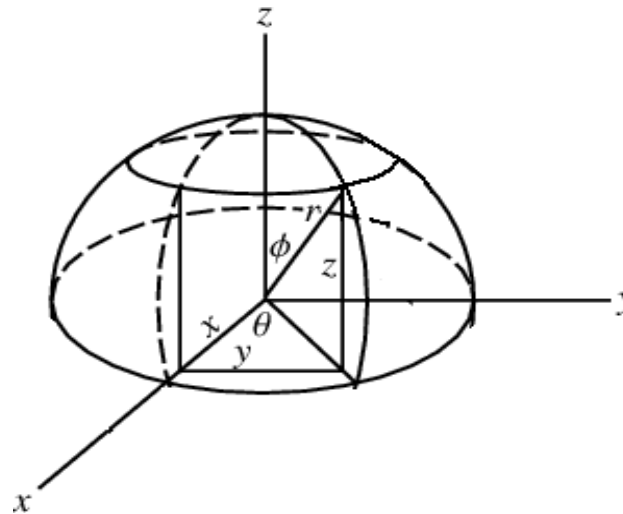


Basics of RADAR

- Radar- RAdio Detection And Ranging
- Transmits microwaves



- Elevation position, ϕ
- Azimuth Position, θ



Application of DSP for Radar Signal Processing

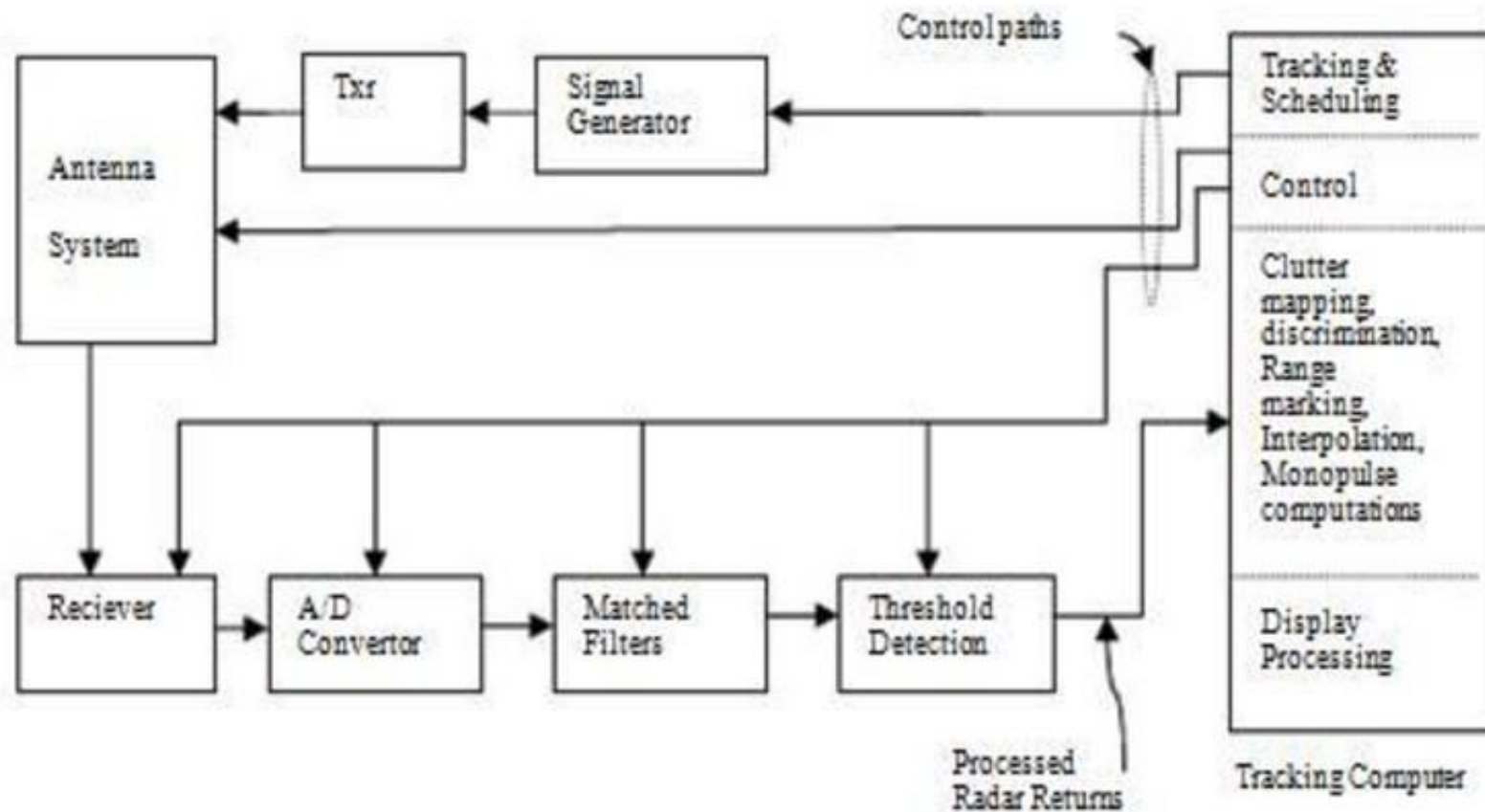


Fig: block diagram of RADAR

Application of DSP for Radar Signal Processing

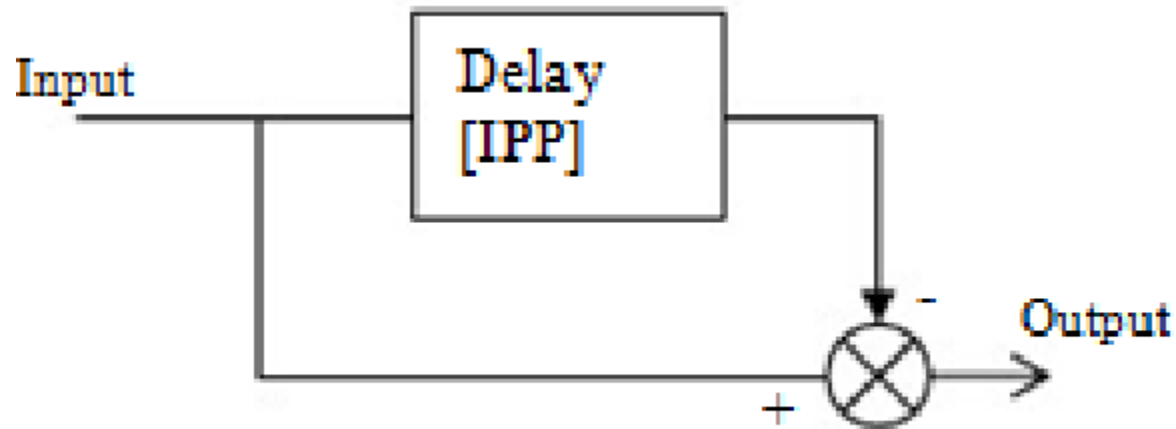


Fig: Cancellor filter

Application of DSP for Radar Signal Processing

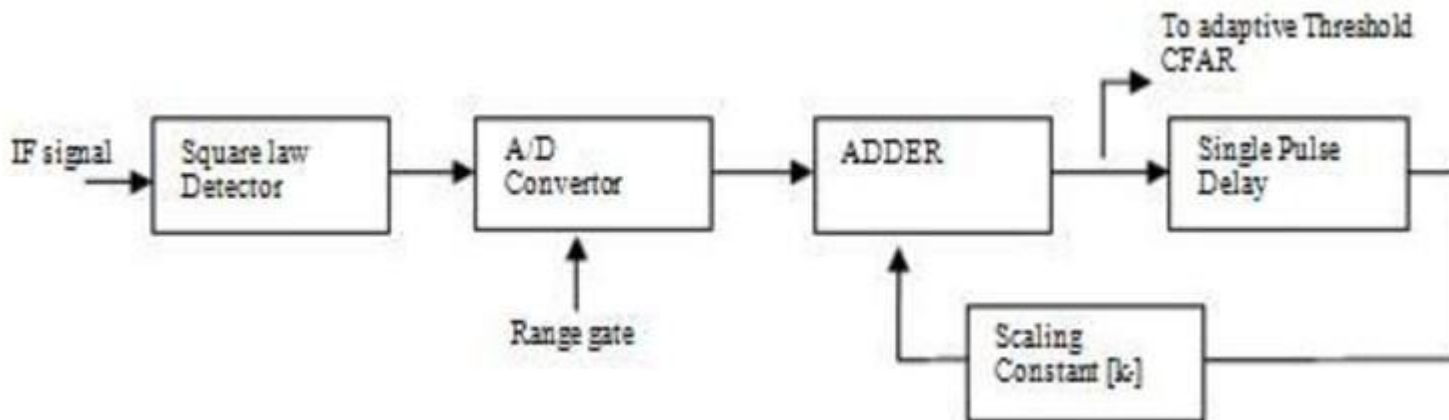


Fig: Square law detector

Application of DSP for Radar Signal Processing

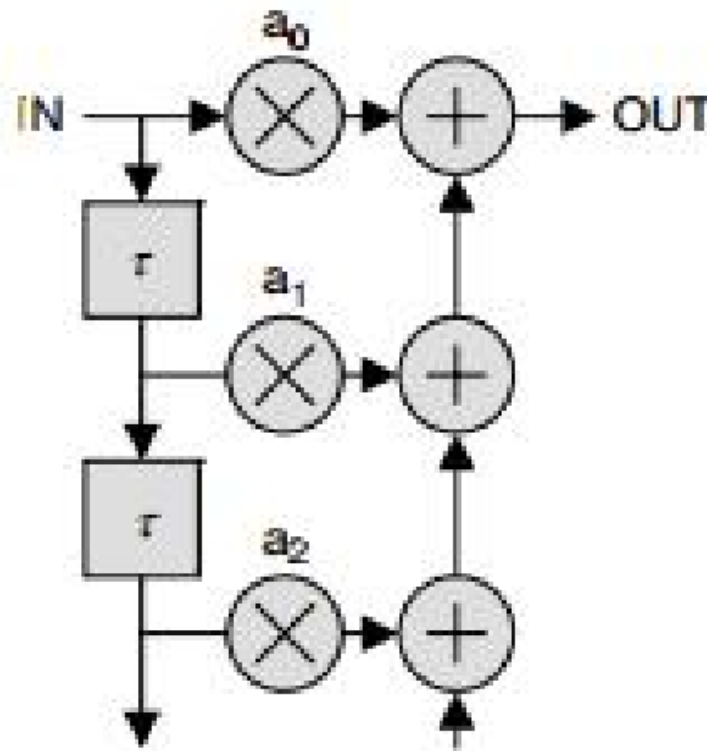


Fig. 4 Block diagram of FIR filter

Thank You